LECTURE 01

MULTIVARIABLE CALCULUS

TODAY'S LECTURE

- Introduction to Calculus
- Reference Axis System
- Concept of Function

Rate of change of dependent w.r.t. independent is called "Differential Calculus".

INTRODUCTION

- Calculus is the mathematical tool used to analyze changes in physical quantities.
- Calculus is also Mathematics of motion and change.
- Where there is motion or growth, where variable forces are at work, producing acceleration, calculus is right mathematics to apply.

DIFFERENTIAL CALCULUS

Deals with the problem of finding

- Rate of change
- Slope of curve

Examples

- Velocities and acceleration of moving bodies
- Firing angles that give cannons their maximum range
- The time when planets would be closest together or farthest apart

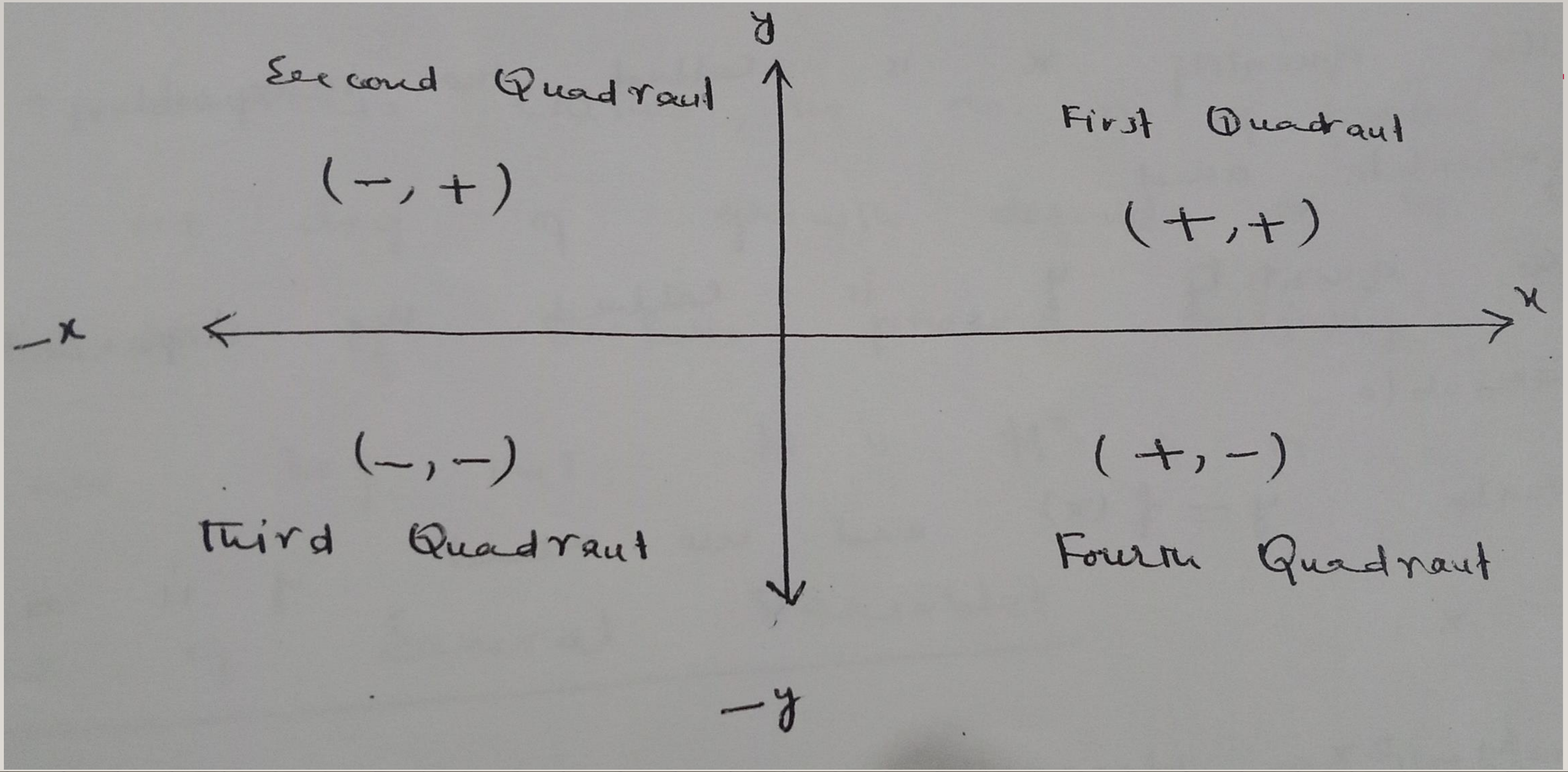
INTEGRAL CALCULUS

Deals with the problem of determining a function from information about its rate of change

Integral Calculus enables us:

- To calculate lengths of curves
- To find areas of irregular region in plane
- To find volumes and masses of arbitrary solids
- To calculate the future location of a body from its present position and knowledge of the forces acting on it

REFERENCE AXIS SYSTEM



FUNCTION

Historically, the term function denotes the dependence of one quantity on another quantity.

Example: The quantity x is called the independent variable and the quantity y is called the dependent variable, we write $y = f(x)$ and we read y is the function of x.

CONT…

The equation $y = 2x$ defines y as a function of x because each value assigned to x determines unique value of y.

Examples of Function:

- The area of circle depends on its radius r by the equation $A = \pi r^2$, so we say that A is a function of r.
- The volume of a cube depends on the length of its side x by the equation $V = x^3$ so we say that V is function of x.

CONT…

- The velocity v of a ball falling freely in the earths gravitational field increases with time t until the ground, so we say that v is function of t.
- In a bacteria culture, the number n of present after one day of growth depends on the number N of bacteria present initially, so we say that N is function of n.

Functions of Several variables:

- The area of a rectangle depends on its length l and width w by the equation $A = lw$ so we say that A is a function of l and w.

CONT…

- The volume of a rectangular box depends on the length l, width w and height h, by the equation $\mathrm{V} = lwh$, so we say that V is a function of l, w and h.
- The area of a triangle depends on its base length l and height h by the equation $A = \frac{1}{2} lh$ so we say that A is a function of l and h.
- The volume V of a right circular cylinder depends on its radius r and height h by the equation $V = \pi r^2 h$ so we say that V is a function of r and h.

EXAMPLE 01

If $f(x) = 2x^2 - 1$ then

$$f(1) = 2(1)^2 - 1 = 1 \qquad when\ x = 1$$
$$f(4) = 2(4)^2 - 1 = 31 \qquad when\ x = 4$$
$$f(-2) = 2(-2)^2 - 1 = 7 \qquad when\ x = -2$$
$$f(t-4) = 2(t-4)^2 - 1 \qquad when\ x = t - 4$$

$$= 2(t^2 + 16 - 8t) - 1$$
$$= 2t^2 + 32 - 16t - 1$$
$$= 2t^2 - 16t + 31$$

EXAMPLE 02

$$f(x,y) = x^2y + 1 \quad then$$
$$f(2,1) = (2)^2(1) + 1 = 5$$
$$f(1,2) = (1)^2(2) + 1 = 3$$
$$f(0,0) = (0)^2(0) + 1 = 1$$
$$f(1,-3) = (1)^2(-3) + 1 = -2$$
$$f(3a,a) = (3a)^2(a) + 1 = 9a^3 + 1$$
$$f(ab,a-b) = (ab)^2(a-b) + 1 = a^3b^2 - a^2b^3 + 1$$

EXAMPLE 03

$f(x,y) = x + \sqrt[3]{xy}$ then

a) $f(2,4) = 2 + \sqrt[3]{(2)(4)} = 2 + \sqrt[3]{8} = 2 + (2)^{3*1/3} = 4$

b) $f(t,t^2) = t + \sqrt[3]{tt^2} = t + \sqrt[3]{t^3} = t + t = 2t$

c) $f(x,x^2) = x + \sqrt[3]{xx^2} = x + \sqrt[3]{x^3} = x + x = 2x$

d) $f(2y^2, 4y) = 2y^2 + \sqrt[3]{2y^2.4y} = 2y^2 + \sqrt[3]{8y^3} = 2y^2 + 2y$

EXAMPLE 04

$f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2}$ then

$$f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \sqrt{1 - (0)^2 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{2}}$$

Example 05

$f(x, y, z) = xy^2z^3 + 3 \ \ then$

$f(2{,}1{,}2) = 2(1)^2(2)^3 + 3 = 19$

$f(0{,}0{,}0) = 0(0)^2(0)^3 + 3 = 3$

$f(a, a, a) = a(a)^2(a)^3 + 3 = a^6 + 3$

$f(t, t^2, -t) = t(t^2)^2(-t)^3 + 3 \ = -t^8 + 3$

$f(-3{,}1{,}1) = (-3)(1)^2(1)^3 + 3 \ = 0$

EXAMPLE 06

$$f(x, y, z) = x^2y^2z^4$$

$$x(t) = t^3, \qquad y(t) = t^2 \quad and \quad z(t) = t$$

$$f(x(t), y(t), z(t)) = \left(x(t)\right)^2\left(y(t)\right)^2(z(t))^4$$

$$= (t^3)^2(t^2)^2(t)^4$$

$$= (t)^6(t)^4(t)^4 = (t)^{14}$$

Example 07

$f(x, y, z) = xyz + x \quad then$

$$f\left(xy, \frac{y}{x}, xz\right) = (xy)\left(\frac{y}{z}\right)(xz) + xy = xy^2z + xy$$

EXAMPLE 08

$$g(x, y, z) = z \sin(xy)$$

$$u(x, y, z) = z^3x^2, \qquad v(x, y, z) = Rxyz, \qquad w(x, y, z) = xy/z$$

Then

$$g(u(x, y, z), v(x, y, z), w(x, y, z)) = w(x, y, z) \sin((u(x, y, z))(v(x, y, z)))$$

$$= \left(\frac{xy}{z}\right) \sin((z^3x^2)(Rxyz))$$

$$= \left(\frac{xy}{z}\right) \sin((\mathrm{R}x^3yz^4)(Rxyz))$$

FUNCTION OF ONE VARIABLE

A function f of one real variable x is a rule that assigns a unique real number $f(x)$ to each point x in some set D of the real line.

Function of two variables

A function f in two real variables x and y is a rule that assigns unique real number $f(x, y)$ to each point (x, y) in some set D of the xy-plane.

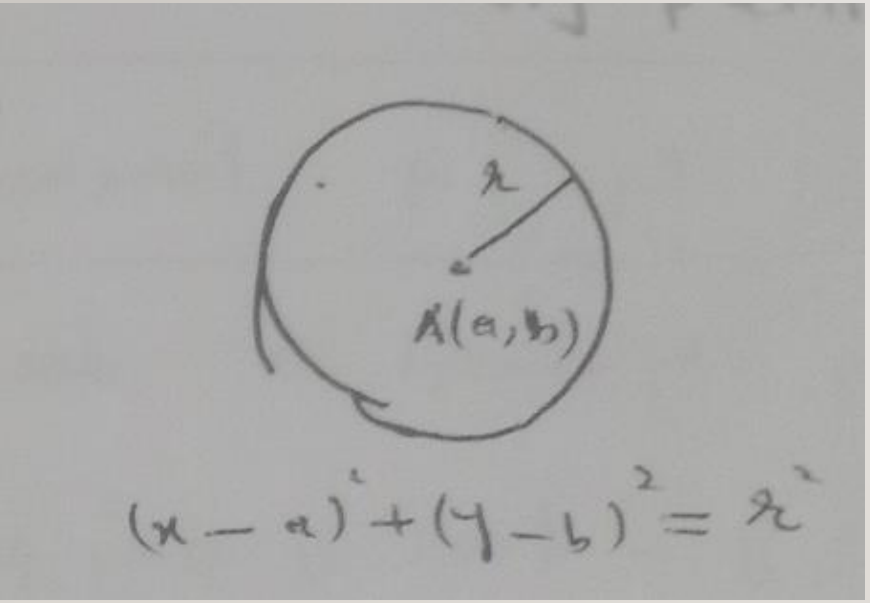
Function of three variables

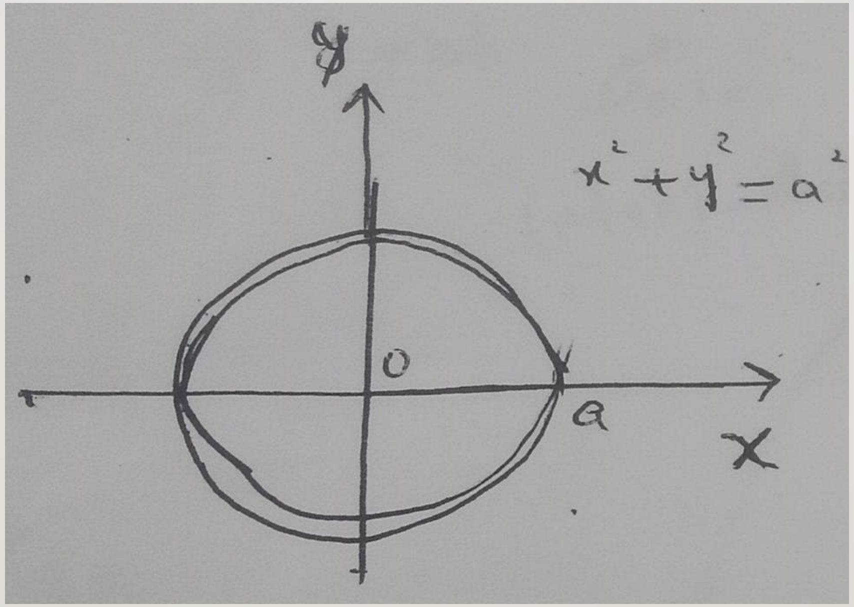
A function f in three real variables x, y and z is a rule that assigns unique real number $f(x, y, z)$ to each point (x, y, z) in some set D of three dimensional space.

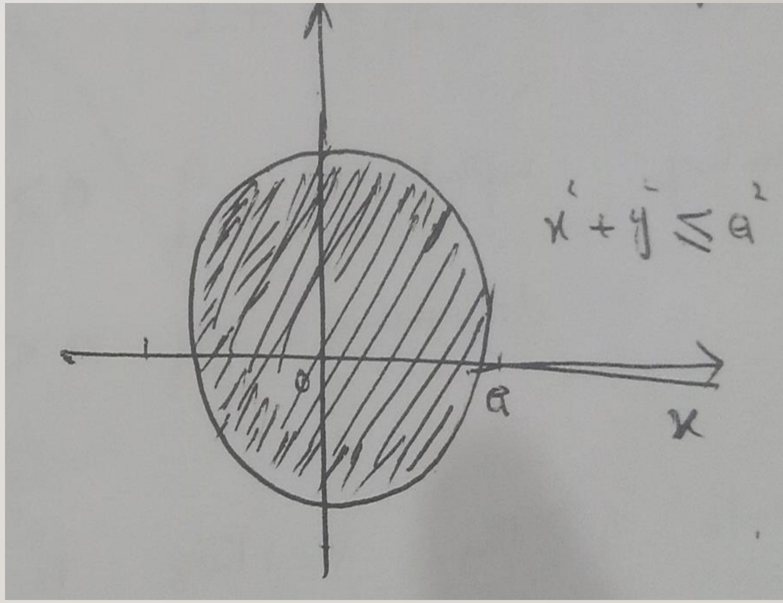
FUNCTION OF N VARIABLES

A function f in n real variables $x_1, x_2, x_3, \dots, x_n$ is a rule that assigns a unique real number $\mathrm{w} = f(x_1, x_2, x_3, \dots, x_n)$ to each point $(x_1, x_2, x_3, \dots, x_n)$ in some set D of n dimensional space.

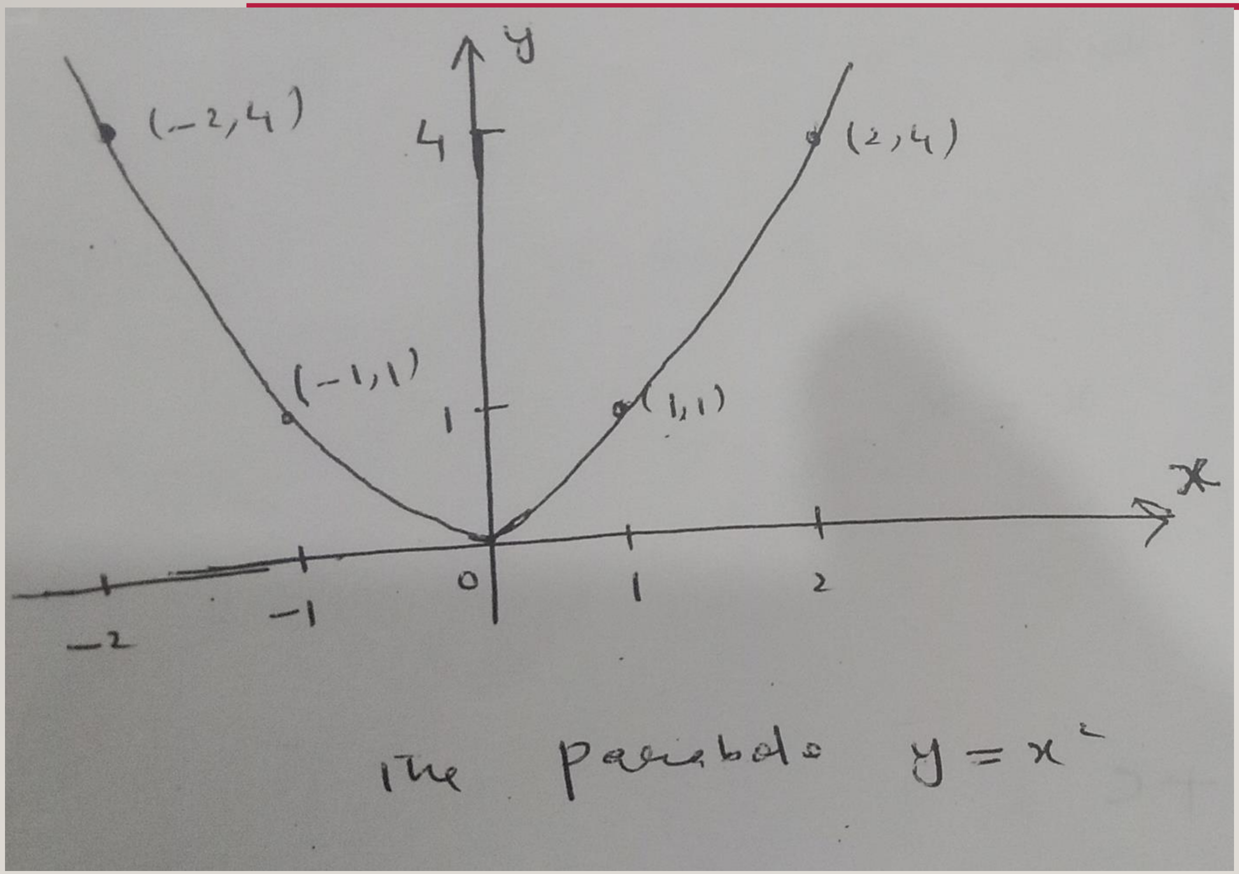
CIRCLE (INTERIOR AND BOUNDARY POINTS OF REGION)



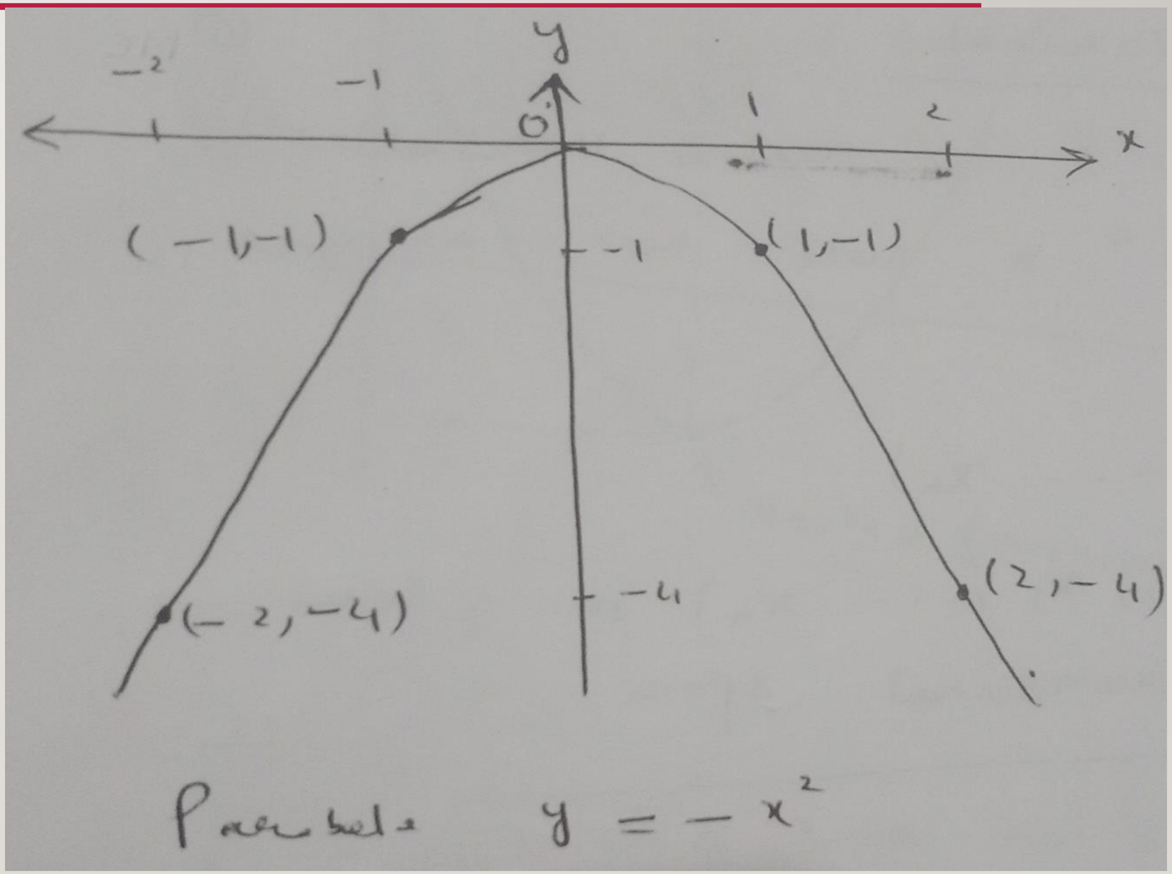




PARABOLA



y
x
(−2,4)
(2,4)
(−1,1)
(1,1)
4
1
0
−2
−1
1
2
The parabola y = x²



y
x
O
−2
−1
1
2
(−1,−1)
(1,−1)
−1
−4
(−2,−4)
(2,−4)
Parabola y = − x²

GENERAL EQUATION OF PARABOLA

Opening upward or downward is of the form

$$y = f(x) = ax^2 + bx + c \qquad a, b, c \epsilon \mathbb{R}$$

Opening upward if $a > 0$

Opening downward if $a < 0$

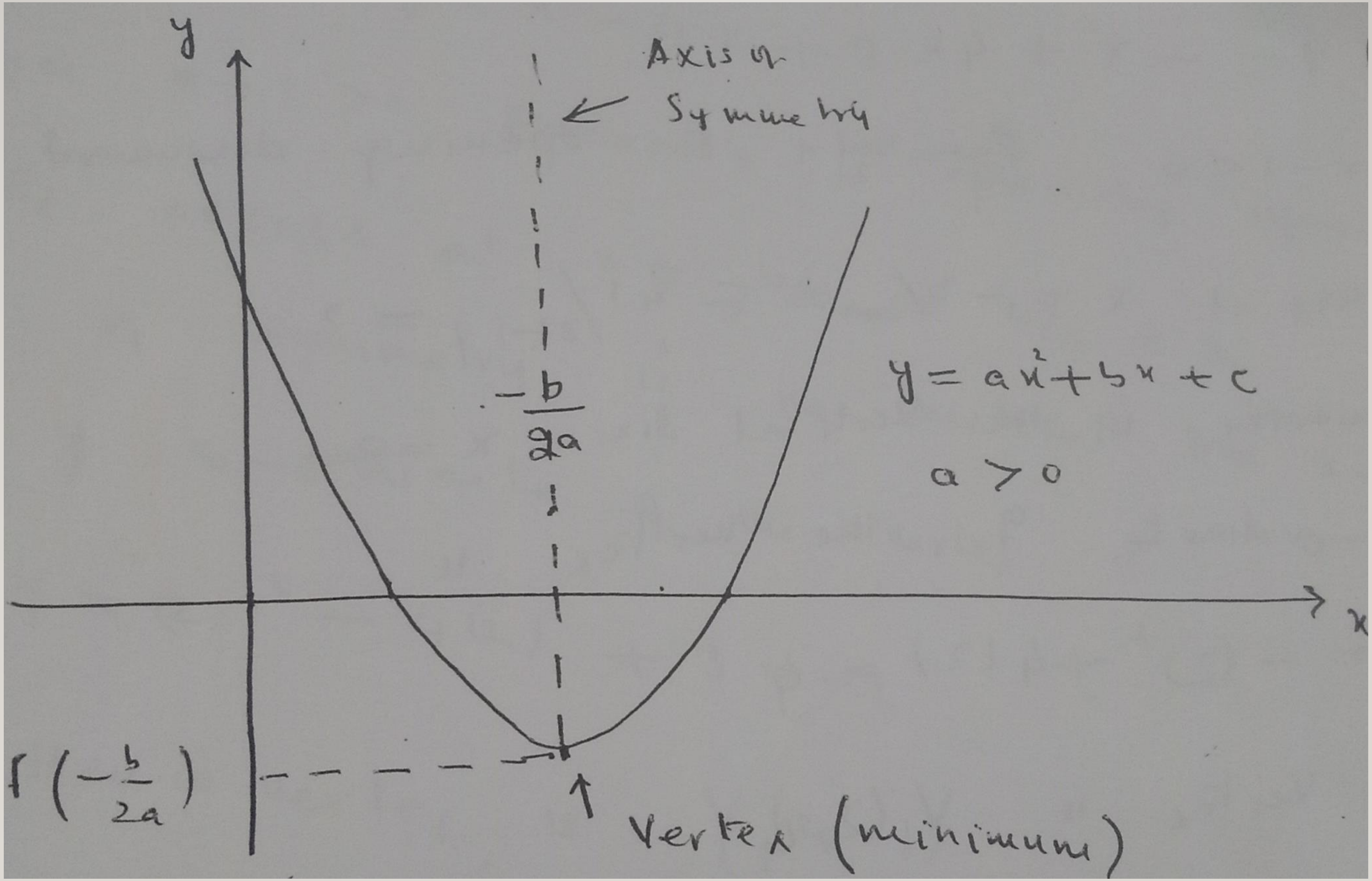
$x - coordinate$ of the vertex is given by $x_0 = -b/2a$

y$-coordinate$ of the vertex is given by $y_0 = f(x_0)$

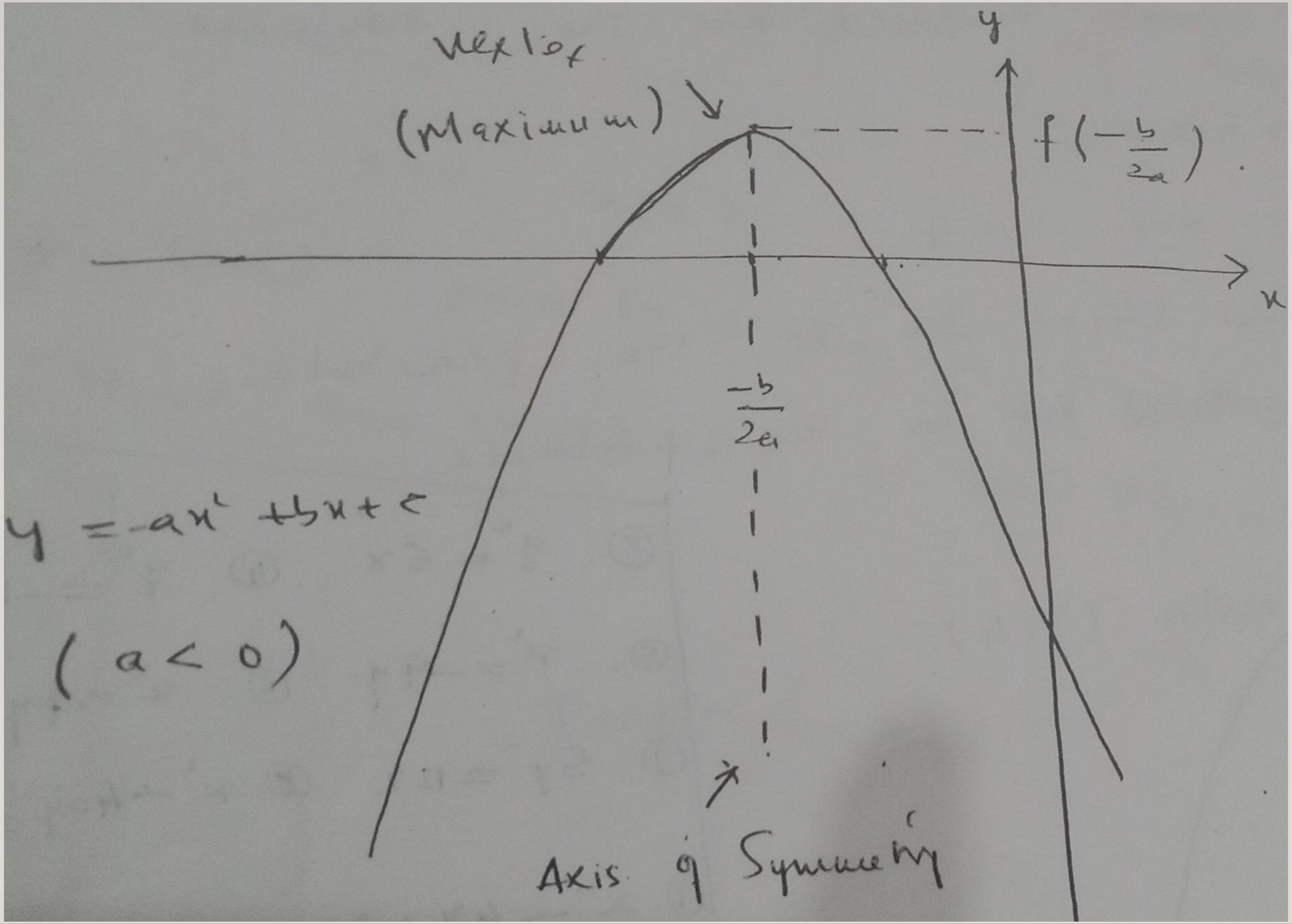
Axis of symmetry is $x = x_0$

Similarly for x-axis $x = ay^2 + by + c$

$$y_0 = -\frac{b}{2a} \qquad and \qquad x_0 = f(y_0)$$



y
Axis of
Symmetry
$-\frac{b}{2a}$
$y = ax^2 + bx + c$
$a > 0$
x
$f\left(-\frac{b}{2a}\right)$
Vertex (minimum)



vertex
(Maximum)
y
$f\left(-\frac{b}{2a}\right)$
x
$\frac{-b}{2a}$
$y = -ax^2 + bx + c$
$(a < 0)$
Axis of Symmetry